

# Part A – Theory Answers

## Statistics & Probability for Loan Default Risk Analysis

This document answers the five Part A theory questions from the practical exam, using the loan/customer dataset context. Each answer includes a plain-language explanation, a worked example, and a supporting chart.

### Q1. Explain Mean, Median, Mode in the context of customer income.

**Mean** – the arithmetic average income: sum of all customers' incomes divided by the number of customers. It is sensitive to extreme values (a few very high earners can pull the mean upward).

**Median** – the middle value once every customer's income is sorted from lowest to highest. It is not affected by outliers, so it better represents a 'typical' customer when income is skewed.

**Mode** – the income value (or income bracket) that occurs most frequently in the dataset. It highlights the most common income level among customers.

**Why it matters for loans:** Income is usually right-skewed (many customers earn a modest, similar amount while a few earn very high incomes). In that case  $\text{Mean} > \text{Median} > \text{Mode}$ , so a bank should look at the median, not just the mean, to understand a typical applicant's affordability.

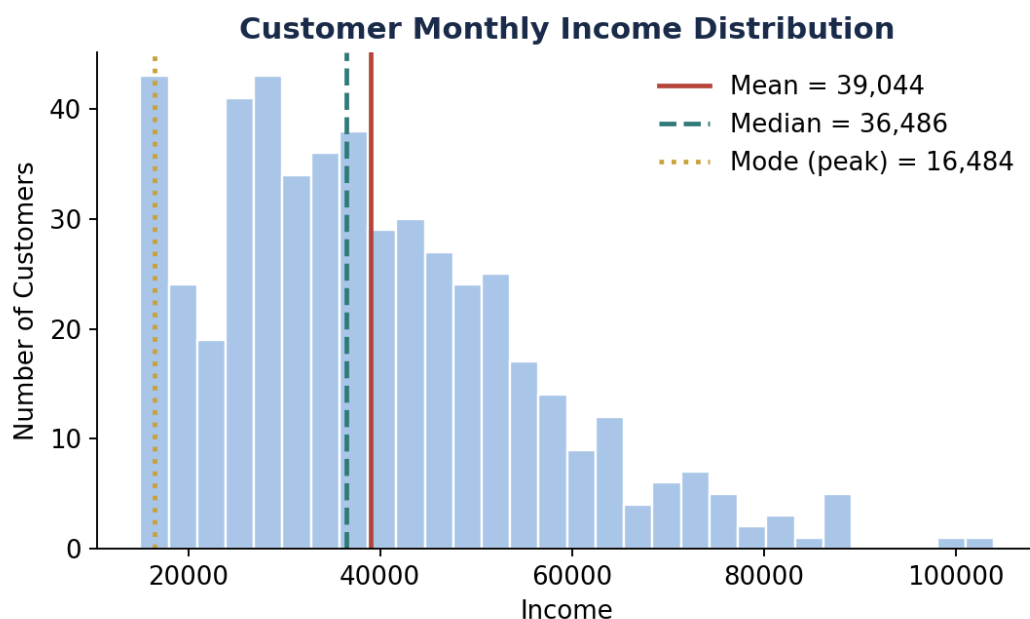


Fig 1. In this skewed income sample, the mean (39,044) sits to the right of the median (36,486), which in turn sits to the right of the mode near the most common income bracket — a classic right-skew pattern.

## Q2. Differentiate between Standard Deviation and Variance using loan amounts.

**Variance** measures how far loan amounts are, on average, from the mean loan amount — but in *squared* units (e.g. rupees-squared), because each deviation from the mean is squared before averaging.

**Standard Deviation** is simply the square root of the variance. Taking the square root brings the measure back into the *original* units (rupees), which is why standard deviation is easier to interpret directly against loan amounts.

Both describe **spread/risk**: a low value means loan amounts cluster tightly around the average (predictable, lower risk); a high value means loan amounts vary widely (harder to predict, higher risk).

### Same Average Loan Amount, Different Spread

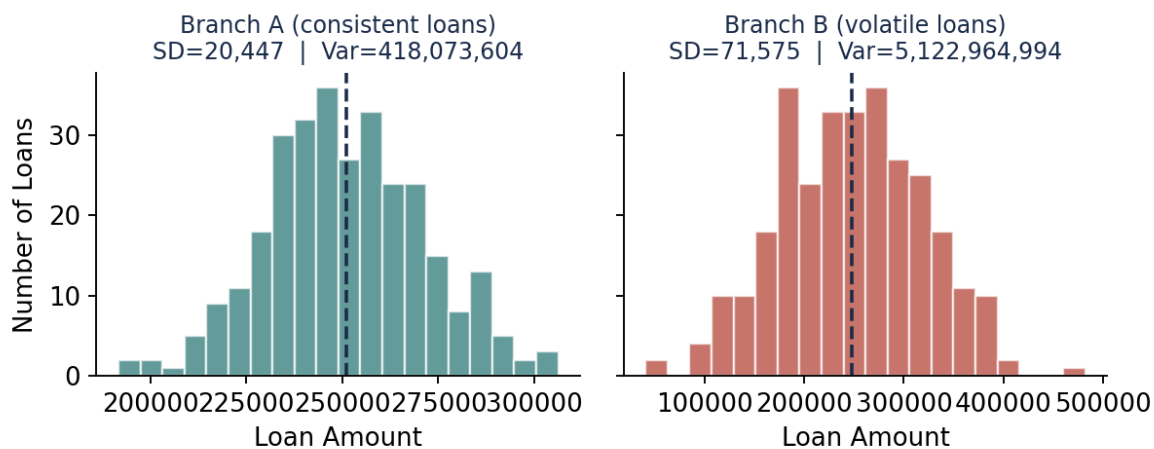


Fig 2. Both branches lend around the same average amount, but Branch B's loans are far more spread out — reflected in its much larger standard deviation and variance.

### Q3. What is a Random Variable? Give one example from the dataset.

A **Random Variable** is a numerical value that comes out of a random or uncertain event, and can change from one observation (e.g. one customer) to the next. It maps each possible outcome of an experiment to a number.

**Discrete random variable** — takes a countable set of values. Example from the dataset: *Loan Default Status* (0 = did not default, 1 = defaulted).

**Continuous random variable** — can take any value within a range. Example from the dataset: *Loan Amount*, which can be any rupee value across a range.

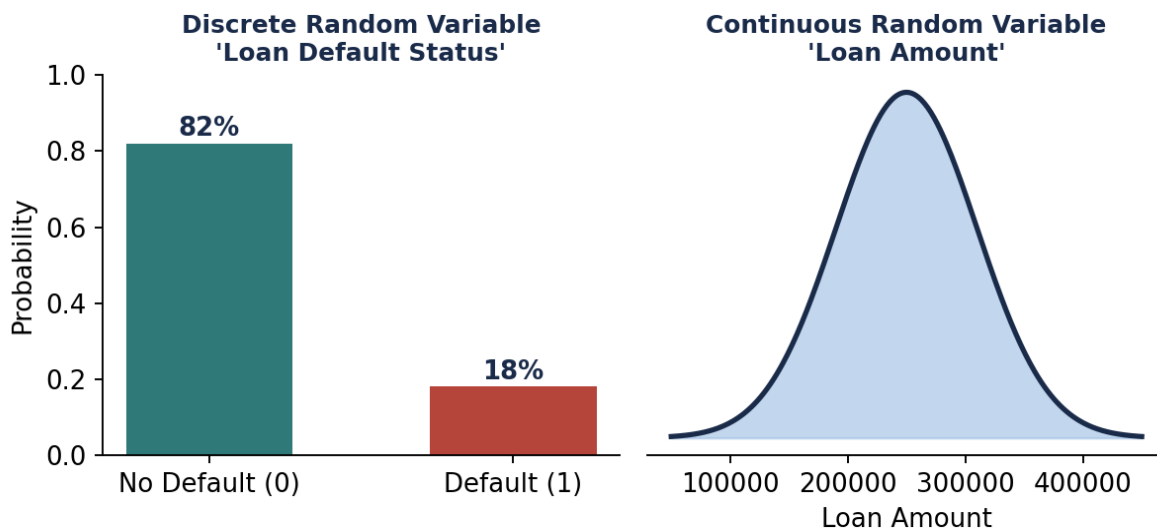


Fig 3. Left: a discrete random variable with two possible outcomes (default / no default). Right: a continuous random variable (loan amount) that can take any value along a range.

#### Q4. Explain Conditional Probability in terms of loan defaults.

Conditional probability is the probability that an event happens **given that another event has already happened**. It is written  $P(A | B)$  — 'the probability of A given B'.

In the loan context:  $P(\text{Default} | \text{Low Credit Score})$  means 'the probability that a customer defaults, given that we already know their credit score is low'. This is far more useful to a bank than the plain probability of default across all customers, because it lets risk be assessed for a specific customer profile.

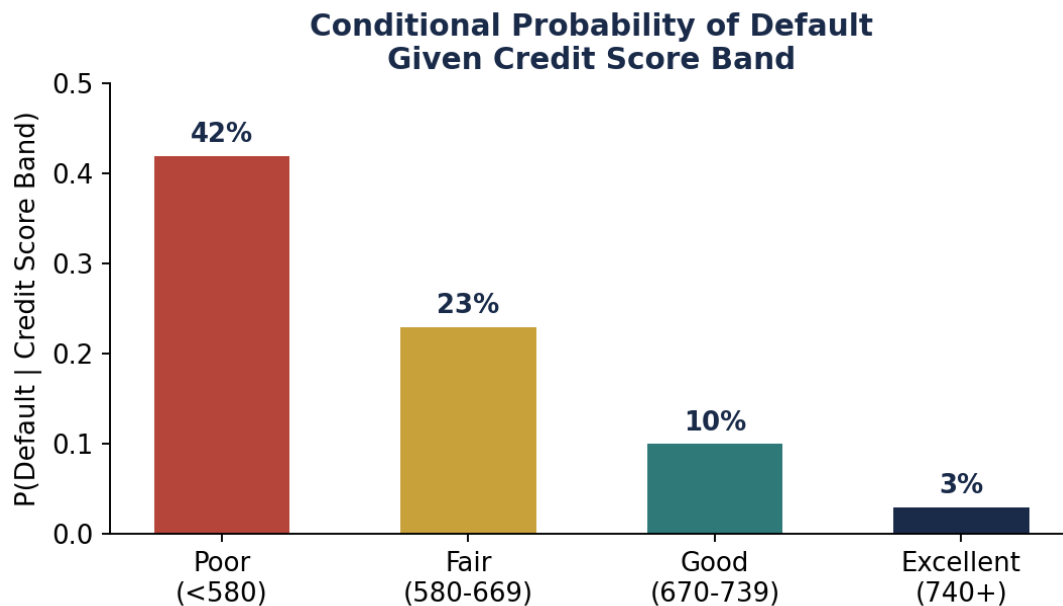


Fig 4. The probability of default changes sharply depending on which credit score band the applicant falls into — this changing probability is exactly what 'conditional' means.

## Q5. Define Bayes Theorem and mention how banks can apply it.

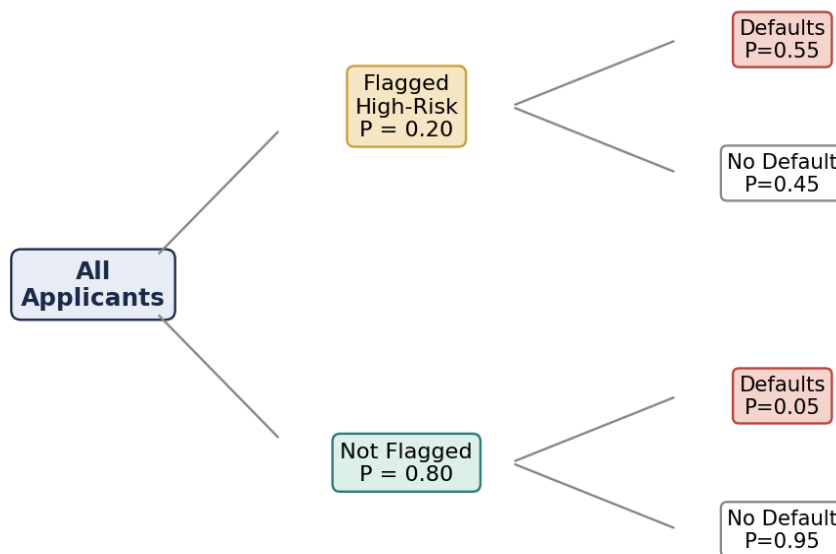
Bayes' Theorem lets us **reverse** a conditional probability — it calculates  $P(A | B)$  when we already know  $P(B | A)$ ,  $P(A)$ , and  $P(B)$ . The formula is:

$$P(A | B) = [ P(B | A) \times P(A) ] / P(B)$$

**Terms:**  $P(A)$  is the prior probability of A before seeing new evidence;  $P(B | A)$  is the likelihood of observing B if A is true;  $P(B)$  is the overall probability of B;  $P(A | B)$  is the updated ('posterior') probability of A once B is known.

**Bank application:** A bank may know  $P(\text{customer defaults})$  and  $P(\text{customer was flagged high-risk} | \text{customer defaulted})$  from historical data. Bayes' Theorem lets the bank compute the reverse and far more useful question:  $P(\text{customer will default} | \text{customer is flagged high-risk today})$ , which directly supports approve/reject decisions on new loan applications.

### Bayes' Theorem: Risk Flag vs. Actual Default



*Bayes Theorem lets the bank flip this around:  
given a customer DID default, what's the probability they were flagged high-risk beforehand?*

Fig 5. The tree shows the 'forward' probabilities the bank already has (flag → default). Bayes' Theorem flips the tree to answer the 'backward' question a loan officer actually cares about: given a default happened, how likely was the risk flag to have caught it?